

Tutorial 06

Registration Number:- 2018/ICTS/47

Name:- Dinuka Dulanjana

Project Description

☐ Login Page

- A user-friendly interface for logging in with a username and password.
- Form validation ensures inputs meet required criteria.
- Successful login redirects users to the Home page.

☐ Registration Page

- Allows new users to sign up by providing details:
 - Full Name
 - Registration Number
 - Index Number
 - Username
 - Email
 - Password and Confirm Password
- Includes form validation techniques to ensure accurate input.
- Provides navigation back to the login page after registration.

☐ Home Page

- Welcomes the user by displaying their username at the top of the page.
- Includes a **Logout Button** for user convenience.
- Displays a list of notes relevant to the logged-in user.
- Provides buttons for navigation:
 - **Add Note Button:** Redirects to the Add Note screen.
 - **View Notes Button:** Redirects to the View Note screen.

☐ Add Note Page

- Allows users to create a new note by entering a title and content.
- Includes a save button to store the note in an array.

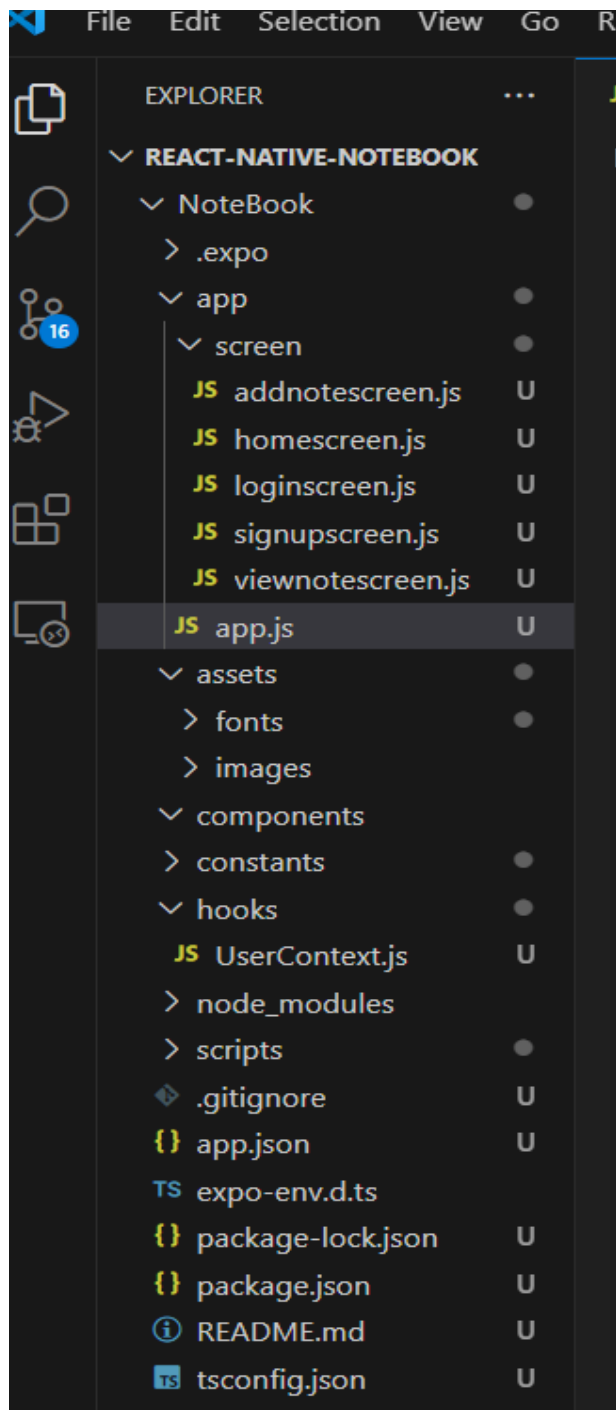
☐ View Note Page

- Displays the full content of a selected note.
- Provides options for managing or editing notes in the future (optional feature).

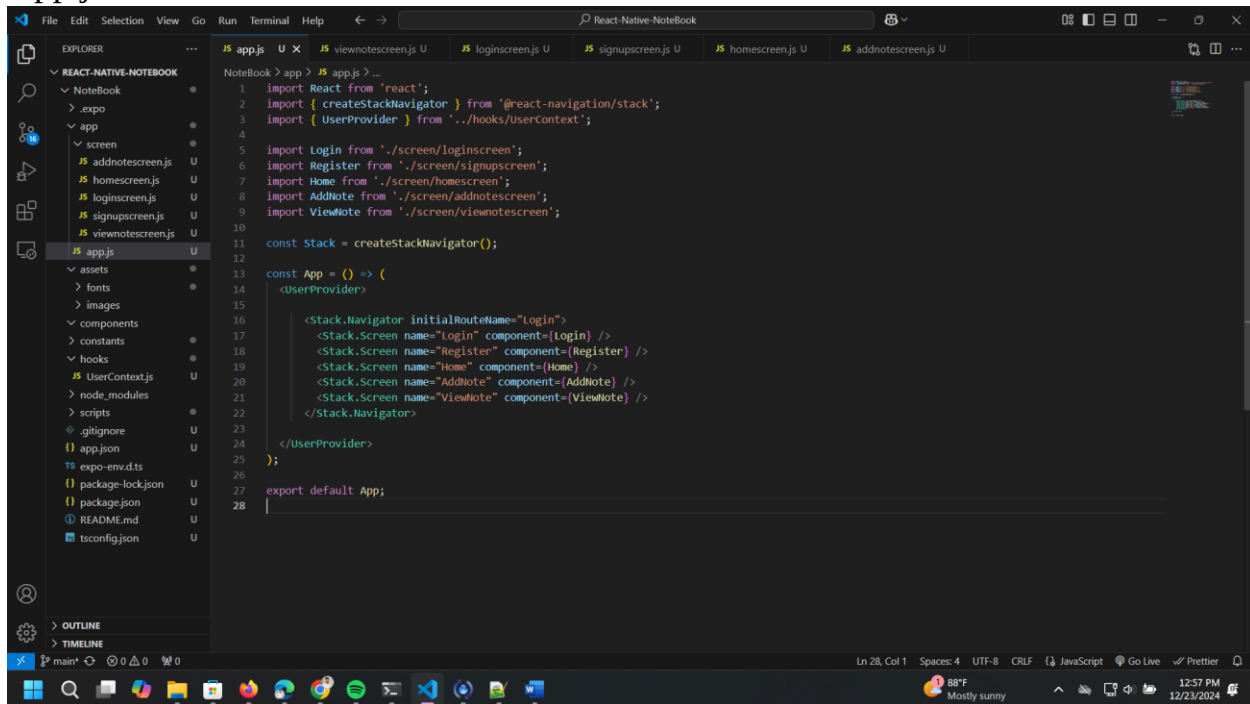
Technical Implementation

- **React Navigation:**
 - Used for seamless navigation between screens.
 - Stack Navigator manages the app's page transitions.
- **Context API:**
 - Centralized state management for user authentication and note data.
- **Form Validation:**
 - Ensures data accuracy on login and registration screens.
- **Data Storage:**
 - Uses an in-memory array to store user details and notes.

File Structure



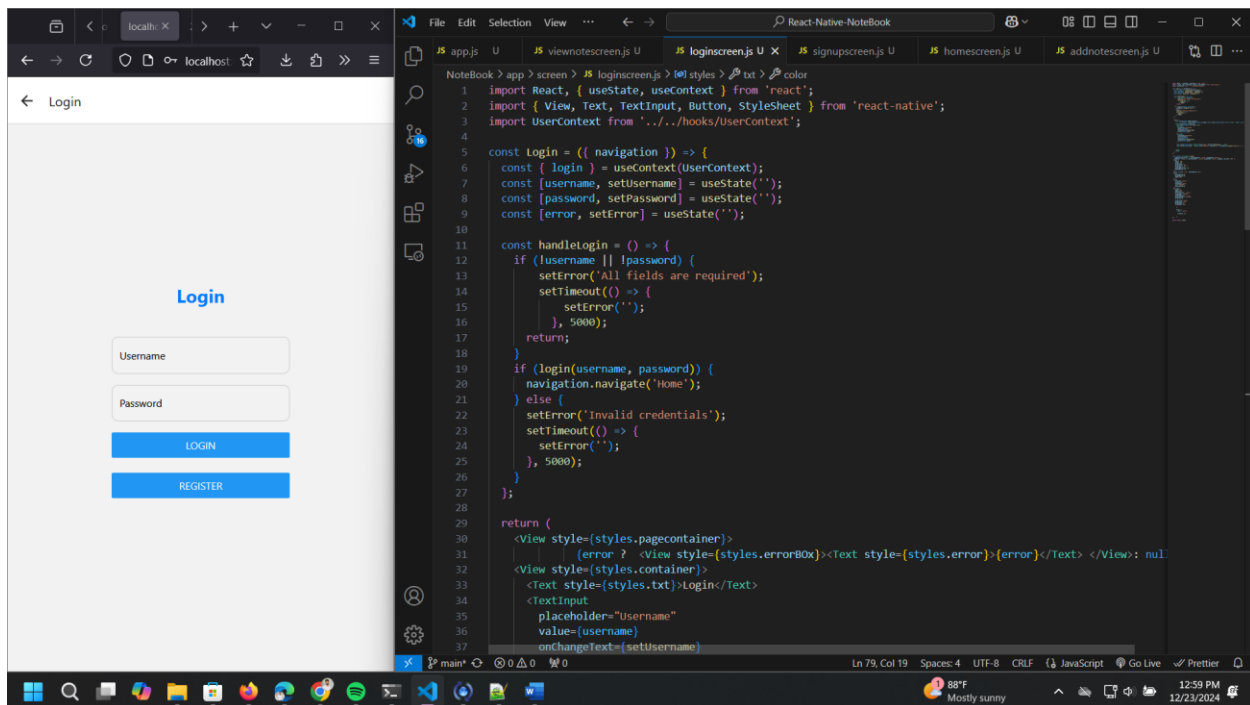
App.js



The screenshot shows the VS Code editor with the 'app.js' file open. The file is located in the 'app' directory of a 'React-Native-Notebook' project. The code defines a navigation stack and a root component 'App'.

```
1 import React from 'react';
2 import { createStackNavigator } from '@react-navigation/stack';
3 import { UserProvider } from '../hooks/UserContext';
4
5 import Login from './screen/loginscreen';
6 import Register from './screen/signupscreen';
7 import Home from './screen/homescreen';
8 import AddNote from './screen/addnotescreen';
9 import ViewNote from './screen/viewnotescreen';
10
11 const Stack = createStackNavigator();
12
13 const App = () => (
14   <UserProvider>
15     <Stack.Navigator initialRouteName="Login">
16       <Stack.Screen name="Login" component={Login} />
17       <Stack.Screen name="Register" component={Register} />
18       <Stack.Screen name="Home" component={Home} />
19       <Stack.Screen name="AddNote" component={AddNote} />
20       <Stack.Screen name="ViewNote" component={ViewNote} />
21     </Stack.Navigator>
22   </UserProvider>
23 );
24
25 export default App;
```

Loginscreen.js



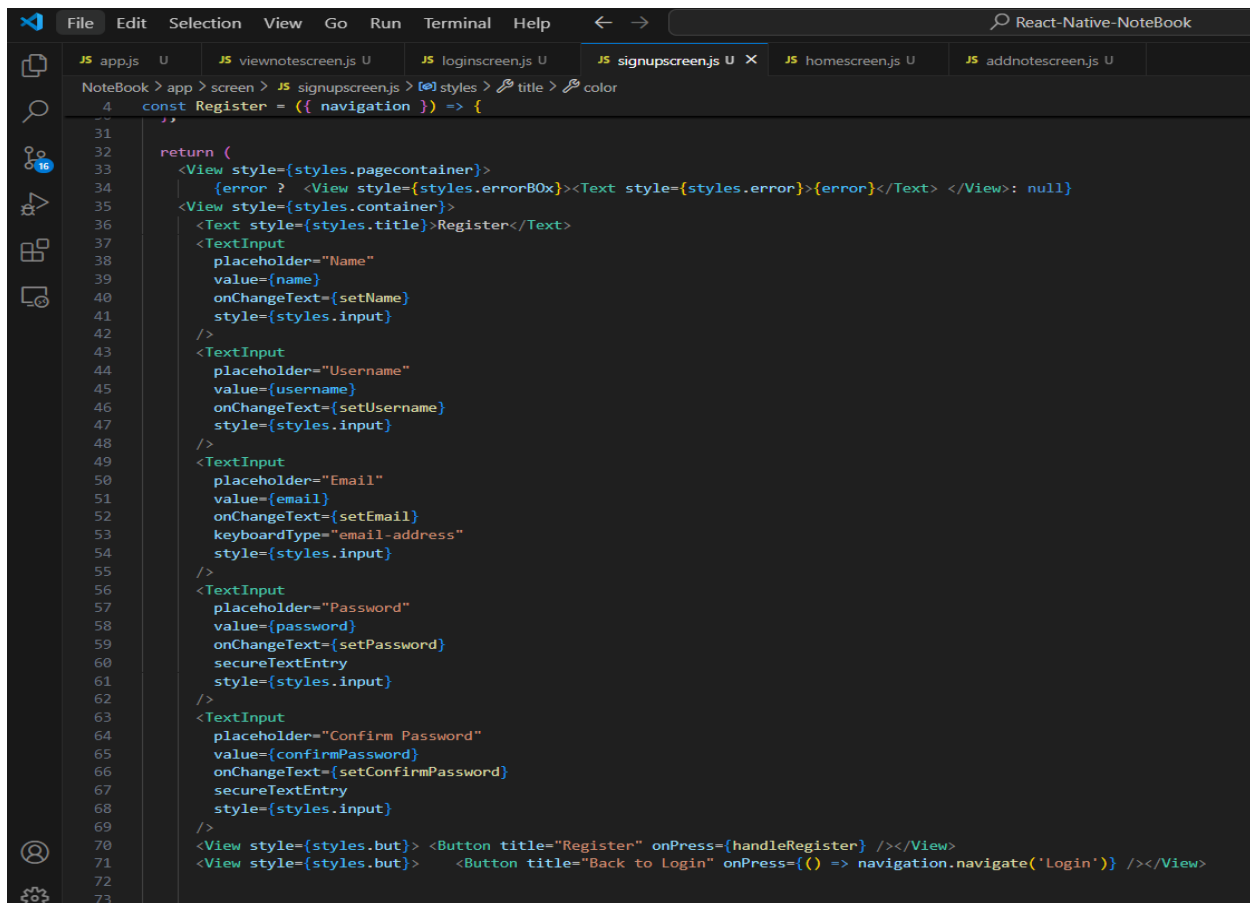
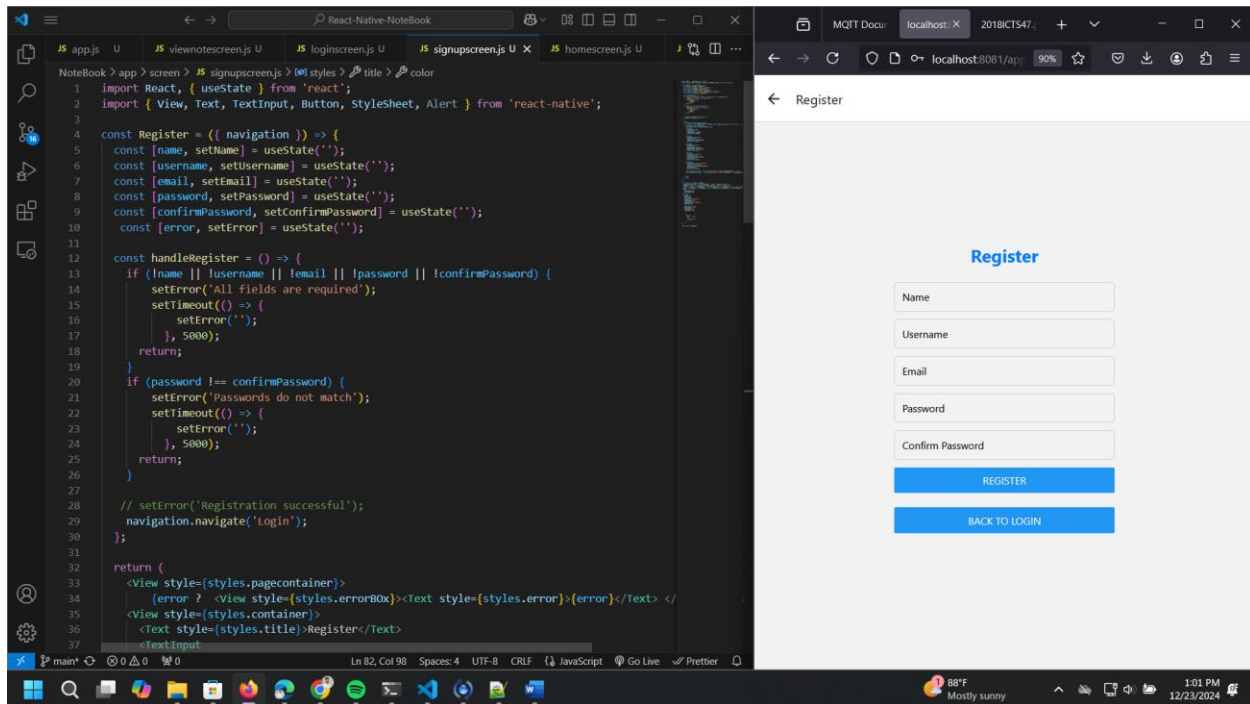
The screenshot shows the VS Code editor with the 'loginscreen.js' file open. The file is located in the 'screen' directory of the 'React-Native-Notebook' project. The code defines a login component that uses the 'UserContext' and 'navigation' prop. It includes a 'handleLogin' function that checks if the username and password are provided and calls 'login' if they are. The UI is rendered using 'View' and 'Text' components, with a 'login' button and a 'register' button.

```
1 import React, { useState, useContext } from 'react';
2 import { View, Text, TextInput, Button, StyleSheet } from 'react-native';
3 import UserContext from '../hooks/UserContext';
4
5 const Login = ({ navigation }) => {
6   const { login } = useContext(UserContext);
7   const [username, setUsername] = useState('');
8   const [password, setPassword] = useState('');
9   const [error, setError] = useState('');
10
11   const handleLogin = () => {
12     if (!username || !password) {
13       setError('All fields are required');
14       setTimeout(() => {
15         setError('');
16       }, 5000);
17       return;
18     }
19     if (login(username, password)) {
20       navigation.navigate('Home');
21     } else {
22       setError('Invalid credentials');
23       setTimeout(() => {
24         setError('');
25       }, 5000);
26     }
27   };
28
29   return (
30     <View style={styles.pagecontainer}>
31       <View style={styles.errorBox}><Text style={styles.error}>{error}</Text> </View>: null
32     <View style={styles.container}>
33       <Text style={styles.txt}>Login</Text>
34       <TextInput
35         placeholder="Username"
36         value={username}
37         onChangeText={setUsername}
38       />
39     </View>
40   );
41 }
```

```
JS app.js U JS viewnotescreen.js U JS loginscreen.js U X JS signupscreen.js U JS homescreen.js U JS addnotescreen.js U
NoteBook > app > screen > JS loginscreen.js > styles > txt > color
5 const Login = ({ navigation }) => {
28
29   return (
30     <View style={styles.pagecontainer}>
31       {error ? <View style={styles.errorBox}><Text style={styles.error}>{error}</Text> </View>: null}
32     <View style={styles.container}>
33       <Text style={styles.txt}>Login</Text>
34       <TextInput
35         placeholder="Username"
36         value={username}
37         onChangeText={setUsername}
38         style={styles.input}
39       />
40       <TextInput
41         placeholder="Password"
42         secureTextEntry
43         value={password}
44         onChangeText={setPassword}
45         style={styles.input}
46       />
47     <View style={styles.but}> <Button title="Login" onPress={handleLogin} /></View>
49     <View style={styles.but}> <Button title="Register" onPress={() => navigation.navigate('Register')} /></View>
50
51   </View>
52 </View>
53
54 );
55 };
```

```
JS app.js U JS viewnotescreen.js U JS loginscreen.js U X JS signupscreen.js U JS homescreen.js U JS addnotescreen.js U
NoteBook > app > screen > JS loginscreen.js > styles > txt > color
56
57 const styles = StyleSheet.create({
58   pagecontainer:{flex: 1, justifyContent: 'center',alignItems:'center' },
59   container: { flex: 1, justifyContent: 'center',alignItems:'center', padding: 20,width:'100%' },
60   input: {
61     width: '50%',
62     height: 50,
63     borderWidth: 1,
64     borderColor: '#ccc',
65     borderRadius: 8,
66     paddingHorizontal: 10,
67     marginBottom: 15,
68   },
69   error: { color: 'red', marginBottom: 10 },
70   but:{
71     marginBottom:20,
72     borderRadius:25,
73     width:'50%'
74   },
75   txt:{
76     textAlign:'center',
77     fontSize:25,
78     fontWeight:'bold',
79     color:'#007BFF',
80     marginBottom:40
81   },
82   errorBox:{
83     width:'50%',
84     padding:15,
85     justifyContent: 'center',
86     alignItems: 'center',
87     borderRadius:5,
88     backgroundColor:'#FF0000',
89     marginTop:20,
90
91     borderStyle:'solid',
92     borderColor:'red',
```

Signupscreen.js



UserContext.js

- **State Management:**

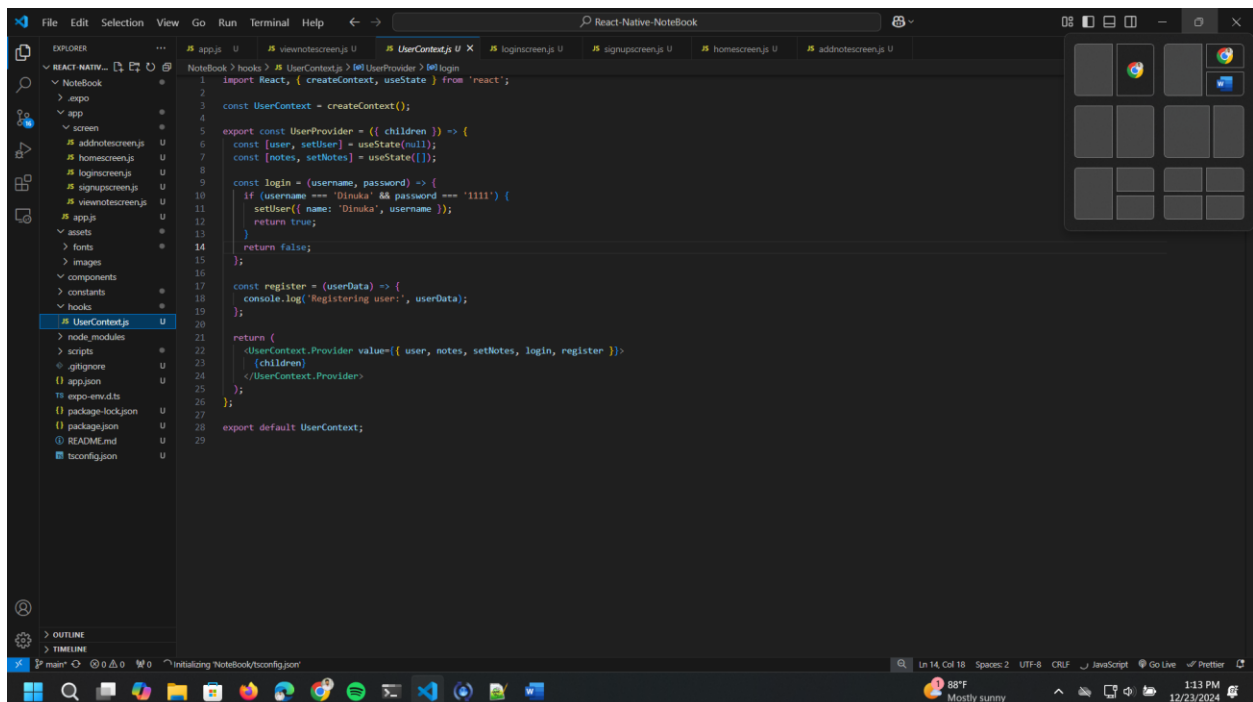
- `user`: Stores the logged-in user's data.
- `notes`: An array of note objects (e.g., `{ title, content, username }`).

- **Context Functions:**

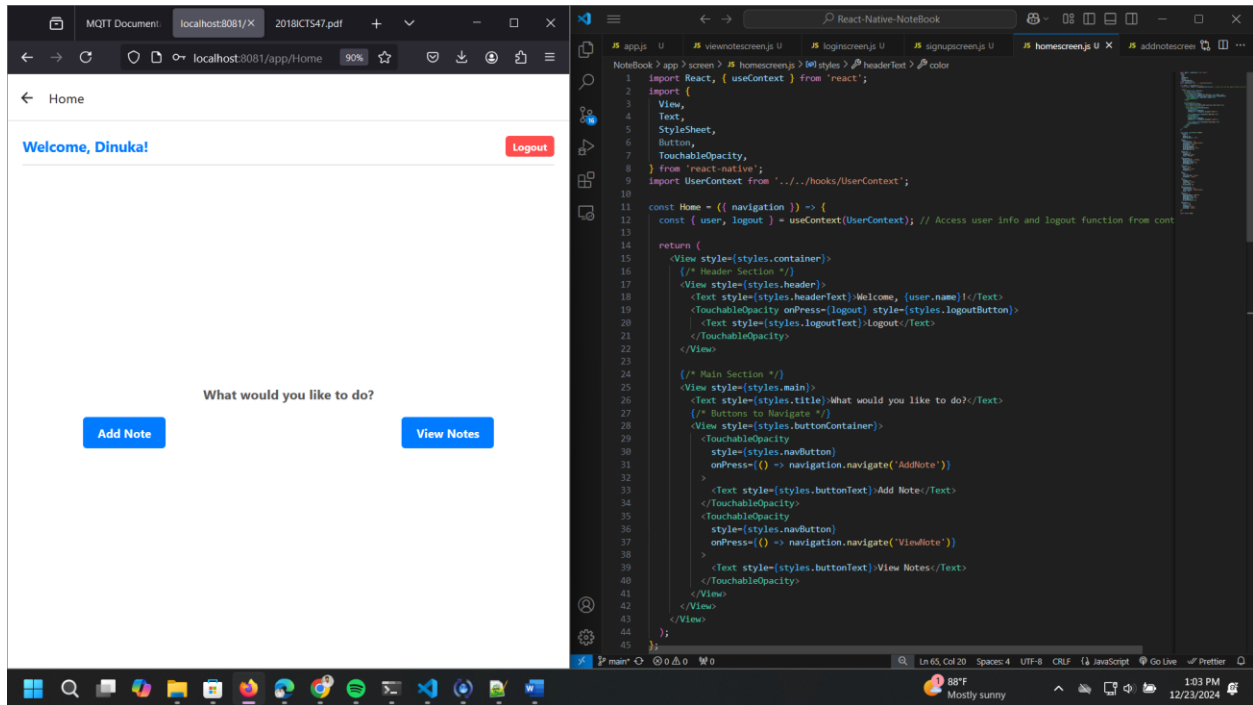
- `loginUser`: Sets the logged-in user's information.
- `logoutUser`: Clears the user's information.
- `addNote`: Adds a new note to the `notes` array.
- `getUserNotes`: Filters and returns notes belonging to the logged-in user.

- **Usage:**

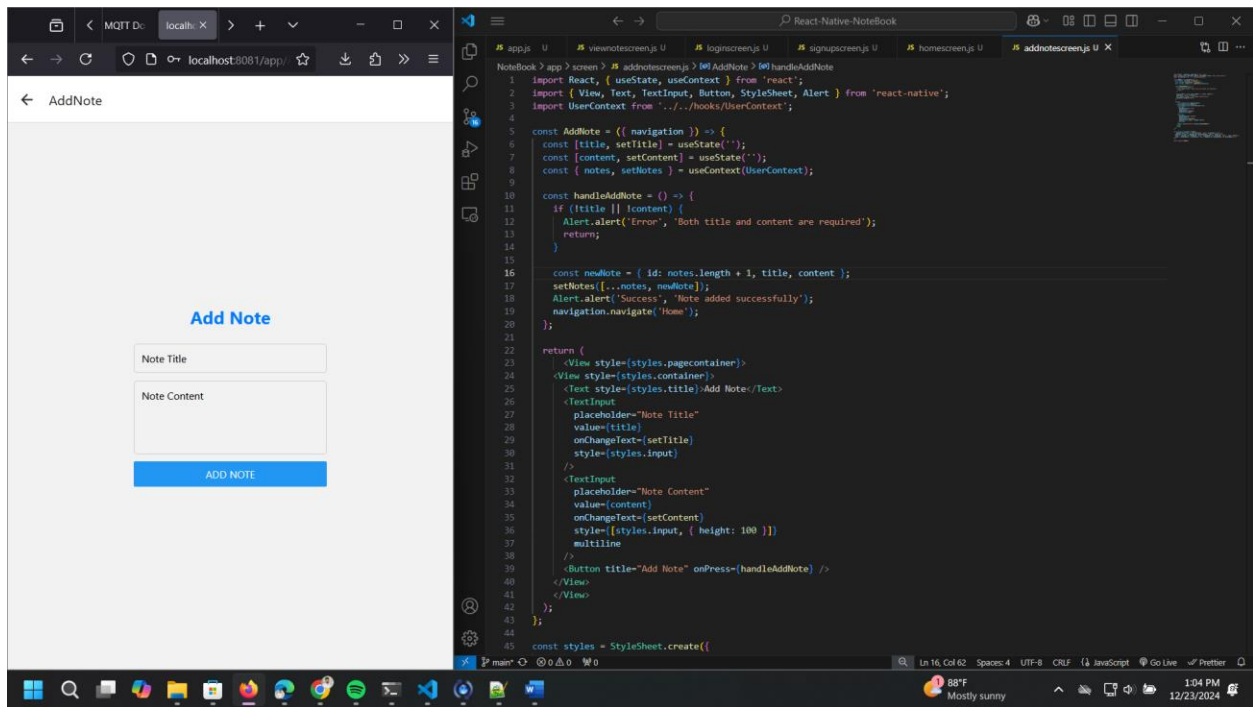
- Wrap the application with `<UserProvider>` to make context data available globally.
- Use `useContext(UserContext)` in components to access or modify user and note data.



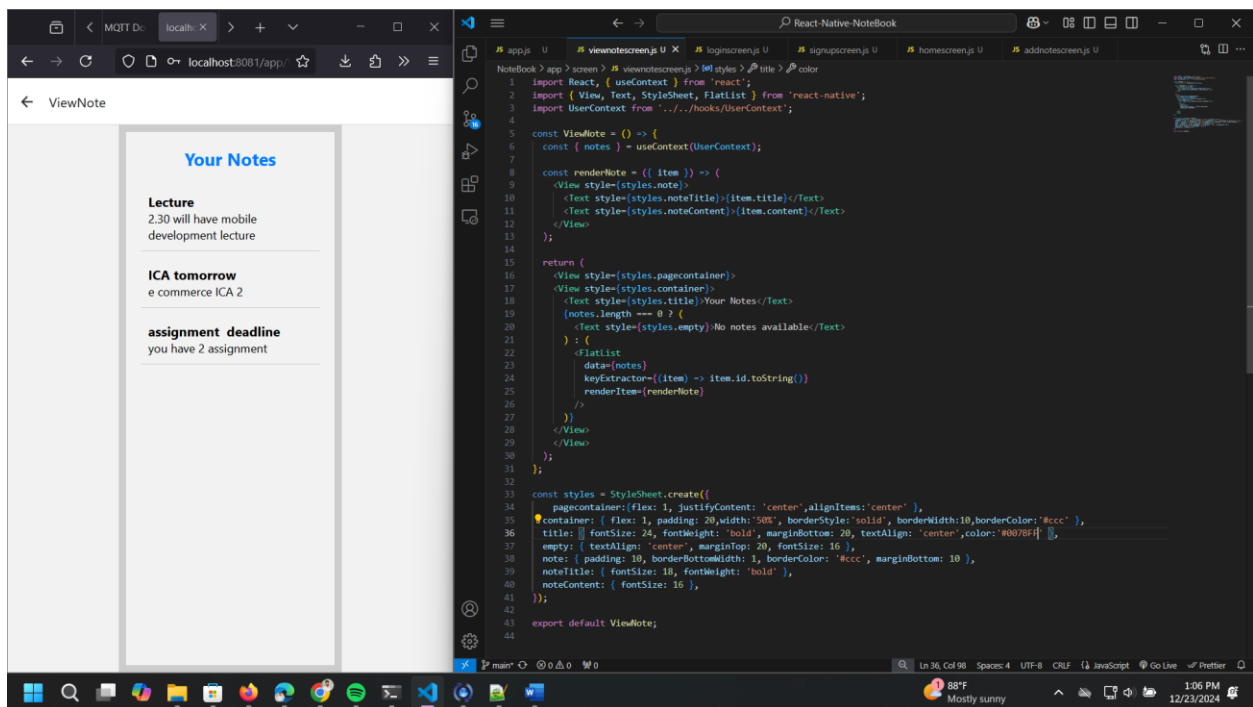
Homescreen.js



Addnotescreen.js



Viewnotscreen.js



Final outputs

MQTT Documentation - MQTT Doc: X localhost:8081/ X 2018ICT547.pdf X +

localhost:8081/app/Register 90% ☆

← Register

Register

Name

Username

Email

Password

Confirm Password

REGISTER

BACK TO LOGIN

Windows taskbar: 88°F Mostly sunny 1:12 PM 12/23/2024

MQTT Documentation - MQTT Doc: X localhost:8081/ X 2018ICT547.pdf X +

localhost:8081/app/Login 90% ☆

← Login

Login

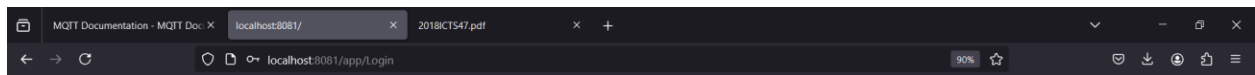
Username

Password

LOGIN

REGISTER

Windows taskbar: Tomorrow's high Near record 1:11 PM 12/23/2024



← Login

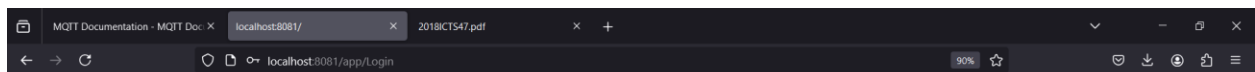
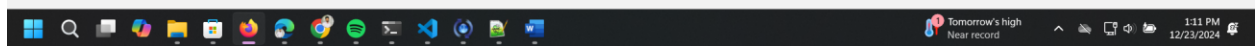
Invalid credentials

Login

dddd

LOGIN

REGISTER



← Login

All fields are required

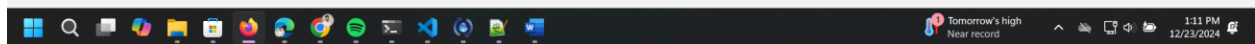
Login

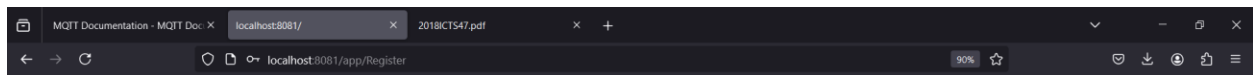
Username

Password

LOGIN

REGISTER





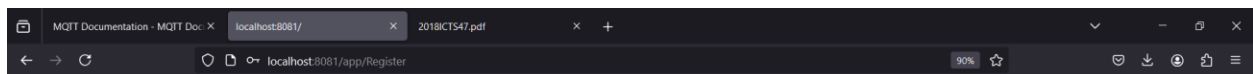
← Register

Passwords do not match

Register

REGISTER

BACK TO LOGIN



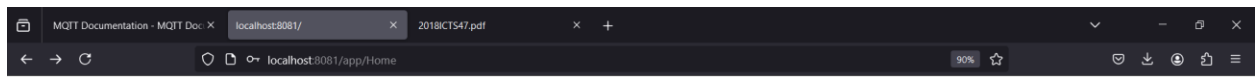
← Register

All fields are required

Register

REGISTER

BACK TO LOGIN



← Home

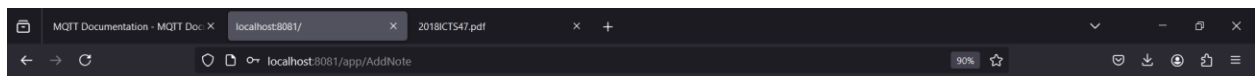
Welcome, Dinuka!

Logout

What would you like to do?

Add Note

View Notes



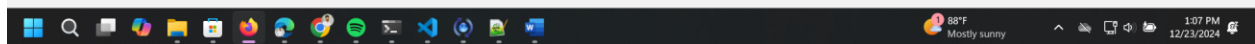
← AddNote

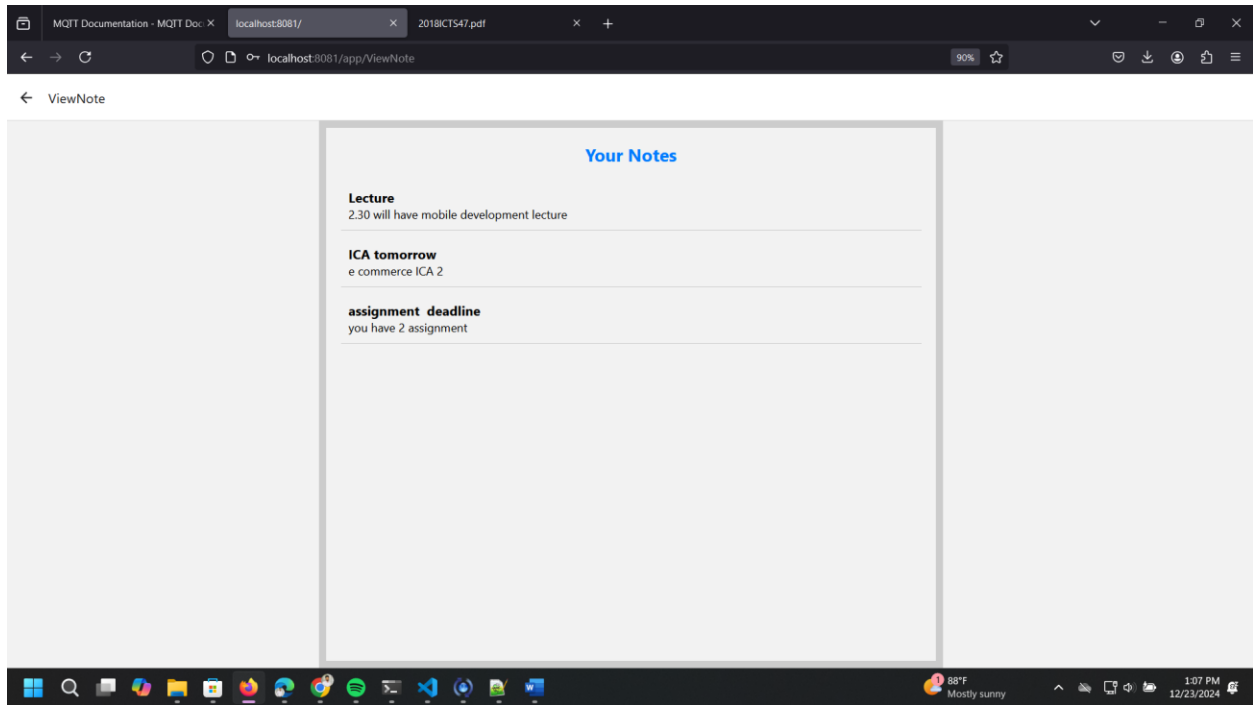
Add Note

Note Title

Note Content

ADD NOTE





Github link:-

<https://github.com/mddranasinghe/React-Native-NoteBook.git>